

Qingkai Dong

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Phone: +1 (860) 931-9075 Location: Storrs, CT, USA

Education

University of Connecticut

Ph.D. in Statistics

Advisors: Prof. HaiYing Wang and Prof. Jun Yan

Aug 2023 – Present

Storrs, CT, USA

Passed the Ph.D. Qualifying Exam

Zhongnan University of Economics and Law

MS in Mathematical Statistics

Thesis: Model Averaging and Variable Selection for Accelerated Failure Time Models

Sep 2020 – May 2023

Wuhan, China

Qingdao University

BS in Applied Statistics

Thesis: Automatic Marking System Based on Convolutional Neural Networks

Sep 2016 – May 2020

Qingdao, China

Research Experience

Research Assistant (Academia–Industry Collaboration)

Servier Pharmaceuticals & University of Connecticut

Mar 2024 – Dec 2025

Coauthored a manuscript introducing a predictive-modeling–assisted interim analysis framework for censored time-to-event endpoints; manuscript in preparation for journal submission.

Developed a covariate-informed approach that augments interim data by predicting event times for censored participants, improving estimation of conditional power and interim decision-making. Implemented in PyTorch.

Proposed evaluation metrics for conditional power forecast accuracy and futility decision quality; conducted extensive simulations to characterize when performance improves or degrades under varied censoring, covariate informativeness/type and prediction range.

Research Assistant (Statistical Methodology)

University of Connecticut

Aug 2023 – Present

Coauthored a methodology paper on rare-feature-aware subsampling for regression models; manuscript in preparation for journal submission.

Developed theory showing that estimation efficiency for rare binary covariate coefficients is driven by the limited number of rare observations, explaining numerical instability and slow convergence.

Introduced balanced subsampling methods that quantifies rarity to ensure adequate representation of rare features without pilot sampling, improving robustness and efficiency.

Implemented the methods in the subsampling R package; validated performance via theoretical results, simulation studies, and real-data applications; produced documentation and reproducible examples.

Research Assistant (Real-World Health Data)

University of Connecticut

Jul 2024 – Aug 2024

Conducted data preprocessing and statistical analyses on All of Us datasets to study associations among social determinants of health, frailty index, and accelerated aging in breast cancer patients.

Delivered analysis-ready datasets and reproducible Python scripts; performed exploratory and confirmatory analyses under missingness and data-quality constraints.

Cross-Disciplinary Collaboration (Chemical Sensor Data Analytics)

University of Connecticut

Mar 2024 – May 2024

Analyzed fluorescence response data from porphyrinoid sensors; applied clustering and statistical analysis to improve compound differentiation and sensor selection.

Produced interpretable summaries and visualizations in R to support experimental decision-making.

Cross-Disciplinary Collaboration (LLM Explainability)

University of Connecticut & collaborators

2025 – Present

Coauthored a survey manuscript on time series explainability emphasizing LLM-enabled semantic explanations; under review at a top survey journal; released an accompanying benchmark repository on GitHub.

Software

R package subsampling: A statistical computing toolkit for optimal subsampling in big-data modeling. Supports GLMs, quantile regression, softmax regression and rare event/rare feature regimes.

Implemented optimized sampling probability computation targeting variance reduction and prediction accuracy, including practical routines for pilot sampling and two-step designs.

Engineering focus: modular API, documentation/vignettes, reproducible examples, and scalable computation patterns.

Teaching Experience

Instructor

University of Connecticut

Jan 2026 – May 2026

STAT 3675Q: Statistical Computing — Comprehensive statistical computing in R/RStudio: data acquisition and management, visualization, reproducible workflows, and core analyses including linear regression/ANOVA, nonparametric inference, GLMs, categorical data analysis, time series, and classification.

Academic Presentations

Poster Presentation - New England Rare Disease Statistics (NERDS) Workshop, Boston, MA 2025

Poster Presentation - Dahshu Data Science Symposium, Storrs, CT 2025

Academic Service

Journal Reviewer: *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

Awards

Predoc Fellowship - University of Connecticut 2025

First-class Scholarship - Zhongnan University of Economics and Law 2020, 2022

Technical Skills

Statistical Methods: survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

Programming: R (package development), Python (machine learning / deep learning), Git, LaTeX

Selected Publications

Dong Q, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.

Zhao H, **Dong Q**. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.

Zhao H, Liu B, **Dong Q**. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.

Chen Z, Lucchesi G, **Dong Q**, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. Manuscript under review at *ACM Computing Surveys*, 2025.